

From residues to feed: detoxification and protein enhancement of agricultural byproducts for animal nutrition

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Abstract

Feed is one of the main costs in animals breeding and several research have been carried out aiming at reducing these costs. An economically interesting alternative is the use of agro-industrial residues in the final formulation of the animal feed. However, many of these residues have properties that may hinder or make it unfeasible for this purpose, such as the presence of toxic or antinutritional compounds or inadequate levels of essential amino acids. In this context, solid-state fermentation (SSF) presents itself as a promising alternative in the use of these residues as a culture medium making its use feasible in animal feed.

This chapter addresses the removal of toxic and antinutritional compounds and protein enrichment of residues through SSF. In addition, the utilization of agro-industrial residues for the production of enzymes by SSF are discussed, for example, xylanases, cellulases, amylases, tannases, and phytases. These enzymes can be used in animal feed in order to improve the bioavailability of certain nutrients.